AIrth

Sustainable Energy

Executive Summary

Data centers are crucial for global digital infrastructure but they are among the fastest-growing sources of carbon emissions. AIrth is an AI-powered digital platform that maps global data centers, predicts their long-term environmental impact, and recommends sustainable actions such as renewable integration, cooling optimization and relocation strategies. The system uses machine learning to forecast future energy demand, grid emissions, and climate risks. AIrth converts complex environmental data into strategic insights that help policymakers and industries accelerate the shift to cleaner digital operations.

The platform contributes directly to Sustainable Energy goals by driving measurable reductions in power use and emissions from global data centers.

Problem Statement

As the world becomes more digital, data centers now power everything from social media to artificial intelligence. Yet this rapid growth carries a major environmental cost. Data centers are among the fastest-growing sources of carbon emissions and electricity use, still heavily reliant on fossil energy and water-intensive cooling systems. According to the International Energy Agency (IEA, 2024), data centers consumed about 460 TWh of electricity in 2022 and could more than double by 2030, putting data centers carbon footprint close to other major industries.

This growing challenge affects data center operators, governments, and researchers, who struggle with limited visibility into where centers are located, how much energy and water they use, and how their footprint may expand in the coming decade.

Problem Statement

1. Context
- Data centers power the digital world but are now one of the fastest-growing sources of carbon emissions and electricity use. Most still rely on fossil energy and water-intensive cooling, making them a major sustainability challenge.

1. 1 Who Is Affected: All struggling with limited visibility into energy use, water consumption, and carbon footprint.

1. 2 User Persona and Pain Point: ex. Bob, a sustainability officer, needs accurate data to track her company's emissions but faces fragmented, outdated, and incomplete.

1. 3 Over 8,000 data centers worldwide consume 1–1.5% of global electricity and could double their demand by 2030, producing up to 300 million tons of CO_2 annually.

2. Research: Our review of IEA, UNEP, and Uptime Institute reports showed that existing tools are regional, self-reported, and lack global predictive capability.

2. 1 User Journey: In recent times, sustainability teams rely on manual reports and scattered data, slowing progress and leading to inefficient, high-emission operations.

Problem Statement

Existing solutions like Google DeepMind’s cooling AI and traditional maps provide partial insights but are regional. There is still no global, integrated system that predicts long-term environmental impacts and supports sustainable digital infrastructure planning.

Without better visibility and forecasting, the environmental footprint of the digital world will continue to expand unchecked. Policymakers will struggle to set effective and global sustainability targets, tech companies will remain reactive rather than proactive in managing emissions. This creates an urgent need for a global, system that not only measures current impact but also anticipates future risks by laying the foundation for sustainable digital transformation.

Proposed Idea - Overview

AIrth uses AI and GIS mapping to forecast data centers long-term environmental footprint and suggest sustainable strategies. The system integrates datasets on location, carbon emissions, energy mix, water use and climate projections (SSP5–8.5) to model future emissions. The overall outcome is a web-based tool that enables sustainable planning for digital infrastructure globally

As for the visualization, panel will contain global map showing data centers and emission hotspots, AI dashboard mockup displaying renewable energy and future risks and recommendation panel suggesting optimal clean-energy strategies.

Proposed Idea - Innovation and Creativity

Current efforts focus solely on specific aspects of data center sustainability. However, they lack a global, integrated approach. AIrth panel is not only tracks emissions but also predicts future impacts and recommends actions such as renewable integration, cooling optimization, and site relocation.

By merging data science with environmental and energy expertise, AIrth transforms sustainability from monitoring to intelligent planning, making digital infrastructure an active part of the clean energy transition.

Proposed Idea - Implementation Plan

1. Deployment: Deployed as a web-based platform accessible to governments, tech companies, and researchers to plan and monitor sustainable data centers worldwide.

a. Milestones: We will start by collecting and cleaning global data, then build AI prediction models, design the user platform, test it with pilot partners, and finally scale it for public access.

b. Resources: Strong technical expertise in AI, GIS mapping, and sustainability analysis, partnerships with data providers and cloud companies, and support from environmental research institutions.

c. Risks: Limited data access, technical complexity, and funding needs, but we will address them through open-data collaborations, modular system design, and partnerships with sustainability-focused organizations

2. Testing: We have run early data modeling tests using sample datasets, which showed strong potential for predicting energy demand and identifying high-emission regions.

3. Financial Overview: The first phase will focus on research and a simple prototype, with estimated costs of $500–$1,000 for software tools, data access, and cloud hosting. Future stages could be supported by grants or academic partnerships.

Proposed Idea - Implementation Plan

Idea: Platform that maps global data centers, predicts their long-term environmental impact, and suggests sustainable solutions.

Phase 1: Research & Concept

- Research of data centers and climate impact
- Mockup and platform concept

Deliverables: Problem statement, concept note

Phase 2: Prototype & Testing

- AI models for emissions forecasting
- Integrate GIS mapping & sustainability data
- Prototype and conduct user testing

Deliverables: Working prototype, testing report

Phase 3: Refinement & Global Impact

- Develop global implementation plan
- Finalize demo and presentation for Global Finale

Deliverables: Final demo, presentation

Proposed Idea - Impact

1. Change and Outcome
- AIrth will help reduce the environmental footprint of global data centers by promoting cleaner energy use, smarter cooling, and better site planning. It will make digital infrastructure more sustainable, efficient, and transparent.

1. 1 Who Benefits: Governments & policymakers – for planning low-carbon infrastructure, tech companies – for optimizing energy and emissions and researchers & NGOs – for more accurate climate insights

1. 2 Expected Impact: Up to 15-25% reduction in energy waste through optimized operation, if this will be not applicable more accurate data report from the data centers will be generated. Lower carbon emissions per data center via renewable integration. Global adoption across facilities.

2. Measuring & Verifying Impact: Remote sensing and GIS data to track site changes and land use. Smart meter/energy reports from partner centers, if collecting data will be not applicable synthetic data will be generated. Lastly, automated analytics dashboard to visualize emissions and energy trends.

3. Success: For us, success means a measurable drop in energy intensity across pilot sites, adoption by industry and government partners, growths of an open database of sustainable data center metrics, and a clear link to reduce CO_2 emissions also smarter energy use.

4. Equity and Inclusion: AIrth is built to be open and accessible to everyone, making sustainability data simple to understand for both experts and the public.

5. Indirect Benefits & Risks: Our project can inspire green jobs, raise awareness about digital sustainability, and bring people together to act on climate issues, while filling data gaps through open collaboration.

Proposed Idea - Impact

Environmental Impact
- Reduction of carbon emissions from data centers.
- Promotes renewable energy use and smart cooling.
- Encourages low-carbon digital growth.

Social Impact
- Helps governments and companies plan sustainably.
- Increases awareness of data center impacts.
- Supports collaboration for greener tech.

Technological Impact
- Uses AI and GIS to forecast environmental risks.
- Turns data into clear, visual insights.
- Builds an open tool for sustainable planning.

Global Impact & SDGs
- SDG 7 – Clean Energy
- SDG 13 – Climate Action
- More resilient digital future

Proposed Idea - Scalability Plan

1. AIrth is designed to scale globally through its data-driven structure. Since it relies on open datasets, it can easily expand to new regions without high infrastructure costs. As more countries and companies publish data on energy grids and emissions, AIrth can integrate these to improve accuracy and local relevance.

2. Because it uses public data sources and cloud-based AI models, scaling requires minimal added cost. As data access improves, accuracy and performance will increase without significant financial growth

3. Early stages can rely on academic or sustainability grants, while later growth can be supported through partnerships with governments, research institutions, and industry sponsors.

4. AIrth aligns with global climate and data transparency goals, supporting initiatives like the EU Green Deal and UN SDG frameworks, while encouraging open environmental data policies.

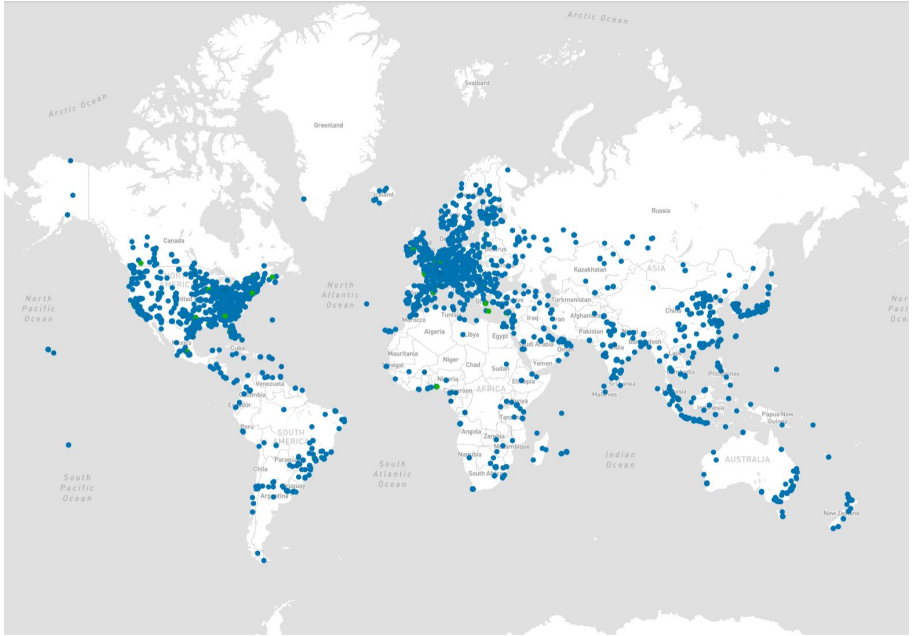
5. In the long run, AIrth will drive lower carbon emissions, energy efficiency, and sustainable digital infrastructure, benefiting both ecosystems and communities through smarter, low-carbon innovation.

Proposed Idea - Supporting Materials

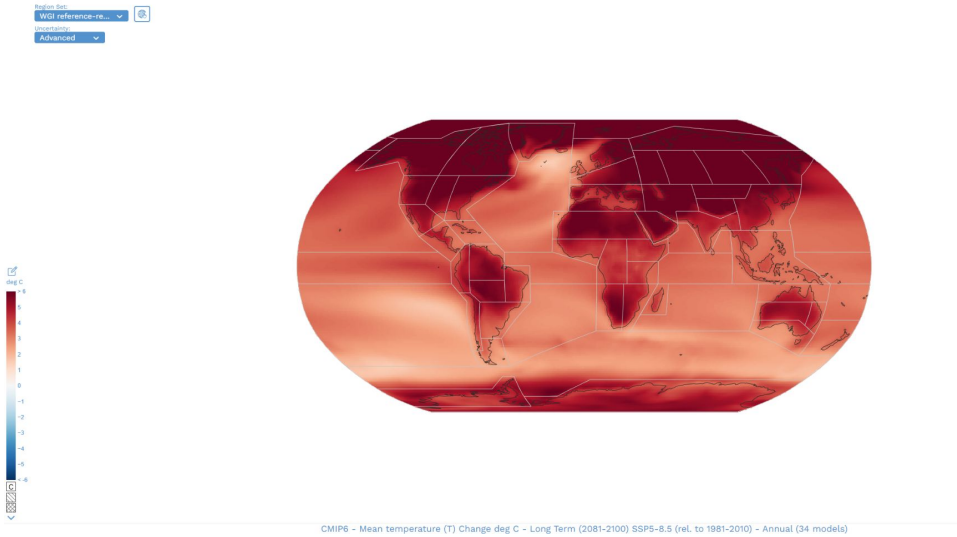
- Currently, the China Telecom–Inner Mongolia Information Park is recognized as the world's largest data center, spanning roughly **1 million square meters**. Its estimated power demand reaches **150 MW**, while its cooling system requires between **12 and 19 million liters of water per day**, highlighting the immense environmental footprint of large-scale digital infrastructure.

- Currently, figures such as these are only estimates, as most data center operators do not publicly share detailed energy or water usage data. With AIrth, these values could be monitored, modeled, and visualized more accurately, allowing governments, companies, and researchers to take timely and informed action. The tool transforms fragmented or hidden data into transparent, evidence-based insights that support sustainable decision-making.

Proposed Idea - Supporting Materials



Data Center Density Map



Global Projected Annual Temp

Comparing global data center density with future temperature projections (SSP5-8.5) shows that rising heat will make current cooling systems increasingly unsustainable. High-density regions will face greater operational stress, highlighting the need for smarter localization, renewable integration, and climate-adaptive cooling strategies.

Proposed Idea - Supporting Materials

- https://www.datacentermap.com (11.7.2025)
- https://www.datacenterdynamics.com/en/news/iea-data-center-energy-consumption-set-to-double-by-2030-to-945twh/ (11.8.2025)
- https://www.iea-4e.org/wp-content/uploads/2025/05/Data-Centre-Energy-Use-Critical-Review-of-Models-and-Results.pdf (11.3.2025)
- https://planet-tracker.org/will-asian-ai-ambitions-be-constrained-by-water-resources/ (11.5.2025)
- https://www.iea.org/commentaries/what-the-data-centre-and-ai-boom-could-mean-for-the-energy-sector (11.5.2025)
- https://climate.copernicus.eu/climate-projections (11.7.2025)
- https://interactive-atlas.ipcc.ch/ (11.8.2025)